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7- List all orders that were delivered later than the estimated delivery date. In your result, include order ID, order estimated delivery date, and actual delivery date.

```
select order_id, order_estimated_delivery_date, order_delivered_timestamp
from orders
where order_delivered_timestamp > order_estimated_delivery_date;
```

	order_id	order_estimated_delivery_date	order_delivered_timestamp
▶	fzKydvN4Nh20	2018-03-02	2018-03-02 23:07:33
	gajqA3huj6aP	2018-05-23	2018-05-25 20:08:38
	MjivV19bQIIN	2018-04-17	2018-04-20 18:12:50
	9R092gnYMHAT	2018-03-12	2018-03-17 02:13:03
	FvGngfs1dIOY	2018-03-26	2018-04-04 15:26:49
	JHskk8RxJONR	2018-03-27	2018-03-27 18:26:42
	JWe1dyIADasJ	2017-12-27	2018-01-05 11:19:40
	C2xtMkYDIZjs	2017-11-03	2017-11-07 00:26:48
	nagCdT81afGS	2018-08-23	2018-09-10 19:53:45
	dQTow9bxPxd1	2018-03-16	2018-04-11 20:43:24
	WPzhW6s5vRdB	2017-07-13	2017-09-19 14:02:52
	vIyf2mS2Lrcp	2018-05-10	2018-05-10 23:12:57
	...	...	...

/*23 rows returned*/

8- Find the total number of customers per state. Sort by the number of customers in descending order.

```
select count(customer_id) as num_customer, customer_state
from customers
group by customer_state
order by num_customer desc;
```

	num_customer	customer_state
▶	199	SP
	60	RJ
	60	MG
	40	RS
	21	SC
	18	PR
	17	GO
	10	DF
	10	PE
	10	BA
	9	ES
	8	MT
	...	...

/* 24 rows returned*/

9- Count the total number of orders and the number of orders with missing shipping charges (NULL).

```
select
    count(order_id) AS total_orders,
```

```
count(order_id) - count(shipping_charges) as missing_shipping_orders
from orderitems;
```

	total_orders	missing_shipping_orders
▶	500	0

/*1 row returned*/

10- For each order status, compute the total number of orders, the earliest purchase date, and the latest purchase date.

```
select
    count(order_id) as num_orders,
    min(order_purchase_timestamp) as earliest_purchase,
    max(order_purchase_timestamp) as latest_purchase,
    order_status
from orders
group by order_status;
```

	num_orders	earliest_purchase	latest_purchase	order_status
▶	347	2016-10-05 17:09:07	2018-08-26 04:52:35	delivered
	151	2017-01-30 23:40:39	2018-08-26 20:32:14	NULL
	1	2017-01-30 21:56:11	2017-01-30 21:56:11	invoiced
	1	2018-05-13 11:11:05	2018-05-13 11:11:05	shipped

/*4 rows returned*/

11- Find the number of orders where the shipping charge is NULL.

```
select count(order_id) as null_shipping_charges_items from orderitems
where shipping_charges is null;
```

	null_shipping_charges_items
▶	0

/*1 row returned*/

12- Find the average product weight and dimensions for each product category and sort by

average weight in descending order.

```
select
    avg(product_weight_g) as avg_weight_g,
    avg(product_length_cm) as avg_length_cm,
    avg(product_height_cm) as avg_height_cm,
    avg(product_width_cm) as avg_width_cm,
    product_category_name
from products
group by product_category_name
order by avg_weight_g desc;
```

	avg_weight_g	avg_length_cm	avg_height_cm	avg_width_cm	product_category_name
▶	10802.5	51.5	56	22	office_furniture
	8350	24	51.5	36.5	luggage_accessories
	7225	16	103	25	industry_commerce_and_business
	6187.5	34.75	21	28.75	NULL
	4971.875	39.125	22	31.25	garden_tools
	4700	35	25	25	costruction_tools_tools
	4467.857142857143	32.857142857142854	30	27.142857142857142	cool_stuff
	4000	19	60	30	small_appliances
	3850	34	24	29	home_confort
	3350	20	21	20	food
	2350	25.5	14.5	22.5	construction_tools_construction
	2111.46875	25.1875	17.5625	20.9375	sports_leisure

/*40 rows returned*/

13- Find the highest-priced order item that does not have a NULL shipping charge. Only shows the top 5 rows and sorts by price in descending order.

```
select
    order_id,
    price,
    shipping_charges
from orderitems
where shipping_charges IS NOT NULL
order by price DESC
limit 5;
```

	order_id	price	shipping_charges
▶	WX3T9ZJET17P	2899	29.05
	AW6Ok89bXKJL	2899	46.76
	Nm193Df52Ekm	2899	37.93
	STFQUGTOWyKD	2799.65	35.95
	FLYZAUFWdOtB	2218	126.47

/* 5 rows returned*/

14- How much time did you spend on this assignment?

45 minutes.